



**Verified Carbon
Standard**

FLOATING SOLAR POWER PROJECT BY AMP ENERGY IN MADHYA PRADESH



Document Prepared by EKI Energy Services Limited



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1 PROJECT DETAILS

1.1 Summary Description of the Project

AMP Energy Green Seven Private Limited, a wholly owned subsidiary of M/s AMP Solar India Private Limited (the Promoter) is a special purpose vehicle incorporated by AMP Energy Green Private Limited to develop a 140 MWp (100 MW_{ac}) floating solar PV plant (the Project) at Omkareshwar water reservoir located near Punasa Tehsil, Khandwa District, Madhya Pradesh State, INDIA. The purpose of this Project is to utilize the area of the reservoir to generate clean energy & supply the same to M.P. Power Management Company Limited (MPPMCL). A long term PPA (Power Purchase Agreement) had been executed between MPPMCL (Procurer) & AMP (Developer).

The project boundary includes the project power plant (project site where the solar power plant has been installed including the solar power plant, power evacuation infrastructure, energy metering points, switch yards and other civil constructions) implemented in the state and all other power plants/units connected physically to the national (Indian) grid that the project power plants is connected to.

The annual average generation of the project is 205,560 MWh. The emission reduction from the project is estimated to be around 191,376 tCO_{2e}/annum and a cumulative emission reduction of 1,339,632 tCO_{2e} for the entire crediting period of 7 years.

Audit Type	Period	Program	VVB Name	Number of years
Validation/ Verification	(31-January-2024- 30-January-2031)	VCS	-	7 Years 00 Month 00 Days
Total	Validation-1	-	-	7

1.2 Sectoral Scope and Project Type

sectoral scope 01: Energy industries¹ (renewable - / non-renewable sources)

Project Type - I

ACM0002 Grid-connected electricity generation from renewable sources --- Version 21.0²

The project is not a grouped project.

¹https://cdm.unfccc.int/DOE/scopes.html?gclid=Cj0KCQiA_P6dBhD1ARIsAAGI7HDggTNy1T26sHqdoxGb1NHqGz8v6FDRT6Dt4%20%20M8vU4uw1i_B-umxt4aAsmDEALw_wcB

² <https://cdm.unfccc.int/methodologies/DB/HF3LP6O41YY0JIP1DK6ZRJO9RSCX3S>

1.3 Project Eligibility

This floating solar project activity involves installation and generation of electricity using renewable energy resources i.e., by using the solar energy (renewable sources) replacing electricity supply from a fossil-fuel dominated electricity, thus leads to reductions of anthropogenic GHG emissions from atmosphere. Hence the project activity is eligible Sectoral scope 1 i.e., energy industries (renewable/ non-renewable sources) under the scope of the VCS Program.

As per section 2.1.3 of VCS Standard ver-4.4, table no. 1 states excluded project activities- (The exclusion does not apply to concentrated solar thermal-to-electricity, floating solar PV, or energy storage systems e.g., batteries).

Thus, the proposed floating solar project is eligible under the scope of the VCS program.

1.4 Project Design

The proposed project has been designed as:

- Single location or installation
- Not a grouped project

Eligibility Criteria

Not Applicable

1.5 Project Proponent

Organization name	AMP Energy Markets India Private Limited
Contact person	Abhishek Pandey
Title	Director
Address	309,3rd Floor, Rectangle One, Behind Sheraton Hotel, Saket, New Delhi-110017
Telephone	+91-9717437676
Email	apandey@ampenergyindia.com

1.6 Other Entities Involved in the Project

Organization name	EKI Energy Services Limited
Role in the project	Project Consultant
Contact person	Mr. Manish Dabkara
Title	MD & CEO
Address	EnKing Embassy, Plot 48, Scheme 78 Part-2, Behind Vrindavan Hotel, Vijay Nagar, Indore-452010, Madhya Pradesh, India
Telephone	+91 9589899649
Email	anil.nayakallu@enkingint.org and registry@enkingint.org

1.7 Ownership

The commissioning certificate for project activity is the supporting document to demonstrate the project ownership. This demonstrates the right of use according to clause 3.7.1 (3) of VCS Standard (v4.4) – “a project ownership arising by virtue of a statutory, property or contractual right in the plant, equipment or process that generates GHG emission reductions and/or removals”. Also, other legal compliances may be considered;

- Consent to Operate
- Commissioning documents
- Consent to Establish

1.8 Project Start Date

Start date of the project activity is 31-January-2024, which is the expected date of commissioning of the project activity

1.9 Project Crediting Period

Crediting Period Start date: 31-January-2024

Crediting Period End date: 30-January-2031

This project activity adopts renewable crediting period of 7 years period which can be renewed for maximum 2 times. This start date of crediting period will vary based on actual commissioning of the project activity.

1.10 Project Scale and Estimated GHG Emission Reductions or Removals

Project Scale	
Project	√
Large project	

Year	Estimated GHG emission reductions or removals (tCO ₂ e)
Year 1	1,98,343
Year 2	1,97,450
Year 3	1,95,673
Year 4	1,93,031
Year 5	1,89,557
Year 6	1,85,292
Year 7	1,80,289
Total estimated ERs	1,459,717
Total number of crediting years	7
Average annual ERs	191,376

1.11 Description of the Project Activity

The proposed 140 MWp (100 MW_{AC}) floating solar PV plant shall be developed on the surface of Omkareshwar water reservoir located near Punasa Tehsil, Khandwa District, Madhya Pradesh State, INDIA. Approximately 589 acres of total water cover is expected to be available for developing the proposed 140 MWp (100 MW_{AC}) floating solar PV plant using ACM0002 ver-21.0³.

The power generated by the floating solar PV plant have evacuated to the utility substation at 110kV. Depending on the quantum of power injected into the substation as recorded by the custody transfer meter, the state utility i.e., MPPMCL would allocate power to the two off takers based on prior agreements with MPPMCL. Wheeling charges shall be borne by the off takers.

³ <https://cdm.unfccc.int/methodologies/DB/HF3LP6O41YY0JIP1DK6ZRJO9RSCX3S>

The proposed project activity is the installation of a new 100 MW_{AC} Solar captive power plant/unit at the site where no renewable power plant was operating prior to the implementation of the project activity (green-field plant⁴)

Details of proposed solar power project is outlined in the table below:

Project Owner	AMP Energy Green Private Limited
Capacity of the Power Plant	140 MWp (100 MW _{AC})
Date of Commissioning	31-January-2024 (Expected)
State	Madhya Pradesh
Connectivity	State Grid

The electricity generated from project activity (operational solar power plant) is connected to the state grid. The project activity thus reduces the anthropogenic emissions of greenhouse gases (GHGs) associated with equivalent amount of electricity generation from the existing grid connected power plants (mostly fossil fuel based) and from addition of new generation sources into the grid. The average annual generation of the project is 205,560 MWh. The emission reduction from the project is estimated to be around 191,376 tCO_{2e} /annum and a cumulative emission reduction of 1,339,632 tCO_{2e} for the entire crediting period of 7 years.

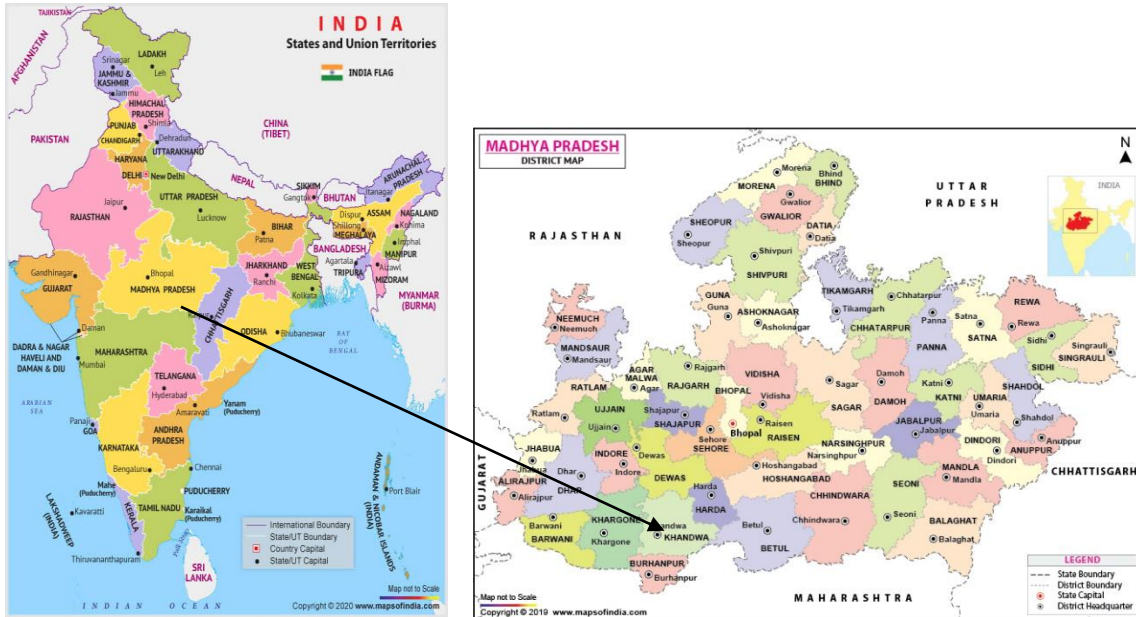
1.12 Project Location

The location of the project activity is-

Village – Omkareshwar
Tehsil – Punasa
District – Khandwa
State – Madhya Pradesh
Country – India

The geographical coordinates are - 22° 11'33.0"N (22.1925), 76° 13'04.0"E (76.2177)

⁴ As per ACM0002 ver20.0, a Greenfield power plant is defined as a new renewable energy power plant that is constructed and operated at a site where no renewable energy power plant was operated prior to the implementation of the project activity.



1.13 Conditions Prior to Project Initiation

There was no project at the site of the project participant prior to the implementation of this project activity. The project participant was not involved in generation of solar PV based power and supplying it to the grid under the pre-project scenario, therefore, in the absence of the proposed project activity, the equivalent amount of electricity would have been generated from the connected / new power plants in the UNIFIED INDIAN GRID, while proposed project activity is a green field project. Therein, the main emission source in the pre-project scenario is the power plants connected to the UNIFIED INDIAN GRID and the primary GHG involved is CO₂.

1.14 Compliance with Laws, Statutes and Other Regulatory Frameworks

The project complies with all Indian relevant laws and regulations. This project is voluntary initiatives by the Project Proponent.

- Electricity Act, 2003 and its subsequent amendment as it does not influence the choice of fuel used for power generation nor has it mandated power generation using renewable energy only.
- Environmental (Protection) Act, 1986 and amendment(s)
- Environmental Impact Assessment (EIA) Notification, 2006 and amendment(s)
- The Air (Prevention and Control of Pollution) Act, 1981 including Rules 1982 and 1983 and amendment(s)
- The Water Prevention and Control of Pollution), Cess Act, 1977 including Rules 1978 and 1991 and amended in March, 2003.
- Noise Pollution (Regulation and Control) Amendment Rules, 2017
- Hazardous and other Wastes (Management & Transboundary Movement) Rules, 2016 and amended in November, 2021.
- E-waste (Management and Handling) Rules, 2020.
- Bio-Medical Waste (Management and Handling) Rules 2016.
- Plastics Waste Management Rules, 2016 amended in 2021.
- Batteries (management and handling) rules 2019.

1.15 Participation under Other GHG Programs

1.15.1 Projects Registered (or seeking registration) under Other GHG Program(s)

The project is not registered nor seeking registration in any other GHG Programs other than VCS.

1.15.2 Projects Rejected by Other GHG Programs

The project is not rejected by any other GHG programs as it has not participated nor seeking registration in any other GHG Programs.

1.16 Other Forms of Credit

1.16.1 Emissions Trading Programs and Other Binding Limits

Net GHG emission reductions or removals generated by the Project will not be used for compliance with an emissions trading program or to meet binding limits on GHG emissions in any Emission Trading program or other binding limits.

1.16.2 Other Forms of Environmental Credit

The project has no intend to generate any other form of GHG-related environmental credit for GHG emission reductions or removals claimed under the VCS program.

Supply Chain (Scope 3) Emissions

Not applicable as the project is not a supply chain project.

1.17 Sustainable Development Contributions

Apart from generation of renewable electricity, the project activity would contribute to the sustainable development of the region - socially, environmentally and economically. Ministry of Environment and Forests, has stipulated economic, social, environment and technological well-being as the four indicators of sustainable development. The project contributes to sustainable development using the following ways

Social well-being: The project would help in generating employment opportunities during the construction and operation phases. The project activity will lead to development in infrastructure in the region like development of roads and also may promote business with improved power generation.

Economic well-being: The project is a clean technology investment in the region, which would not have been taken place in the absence of the VCS benefits the project activity will also help to reduce the demand supply gap in the state. The project activity creates local employment generation which helps economic well-being of local people.

Technological well-being: The successful operation of project activity would lead to promotion of floating solar based power generation and would encourage other entrepreneurs to participate in similar projects.

Environmental well-being: Solar being a renewable source of energy, it reduces the dependence on fossil fuels and conserves natural resources which are on the verge of depletion. Due to its zero emission the Project activity also helps in avoiding significant amount of GHG emissions.

1.18 Additional Information Relevant to the Project

Leakage Management

Not Applicable as the project is not a AFOLU project.

Commercially Sensitive Information

No commercial sensitive information has been excluded from the public version of the project description.

Further Information

There are no information or incidents that will have bearing on the eligibility of the project, the net GHG emission reduction or removals, or the quantification of the project's net GHG emission reductions or removals.

2 SAFEGUARDS

2.1 No Net Harm

There is no harm identified from the project and hence no mitigation measures are applicable.

2.2 Local Stakeholder Consultation

Local stakeholders were invited by sending invitation letters. The consultation was carried out within plant premises.

The local stakeholder consultation process was carried out through a physical meeting at the project site. As a part of the consultation process, the project representatives explained all stakeholders about the type of development through the project and the technology used for power generation towards enhancing stakeholders' awareness about the project. During the meeting the project representative talked about the employment opportunities and other benefits likely to be rendered through this project and its contribution towards fighting against climate change.

The project owner also shared a feedback form during individual interviews during focused group discussions where the stakeholder can put down his/her complaint and the same if found genuine were to be addressed immediately. Also, regular stakeholder engagement is one the key focus at the site.

The project stakeholders were totally in support of setting up of these kinds of projects in the region. All the supporting documents will be provided to VVB during the validation process.

2.3 Environmental Impact

The guidelines on Environmental Impact Assessment have been published by Ministry of Environment, Forests and Climate Change (MoEFCC), Government of India (GOI) under Environmental Impact Assessment notification 14/09/2006⁵. Further amendments to the notification have been done on 14/07/2018⁶. As per the notification:

The following projects or activities shall require prior environmental clearance from the concerned regulatory authority, which shall hereinafter referred to be as the Central Government in the Ministry of Environment and Forests for matters falling under Category 'A' in the Schedule and at State level the State Environment Impact Assessment Authority (SEIAA) for matters falling under Category 'B' in the said Schedule, before any construction work, or preparation of land by the project management except for securing the land, is started on the project or activity:

- (i) All new projects or activities listed in the Schedule to this notification;
- (ii) Expansion and modernization of existing projects or activities listed in the Schedule to this notification with addition of capacity beyond the limits specified for the concerned sector, that is, projects or activities which cross the threshold limits given in the Schedule, after expansion or modernization;
- (iii) Any change in product - mix in an existing manufacturing unit included in Schedule beyond the specified range."

As the solar power generation projects are not listed in any of the categories of the schedule. So, the project is considered environmentally safe and as per Host party- India, no EIA is required.

2.4 Public Comments

Not Applicable

2.5 AFOLU-Specific Safeguards

As the project activity is non-AFOLU this section is not Applicable.

⁵ <http://www.environmentwb.gov.in/pdf/EIA%20Notification,%202006.pdf>

⁶ <http://www.egcipl.com/Doc/Gazette%20Notification.pdf>

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

For Methodology

Title - Grid-connected electricity generation from renewable sources

Reference – ACM0002

Version number – 21.0⁷

The applied methodology draws upon the following tools:

1. Tool 01 - Tool for the demonstration and assessment of additionality (Version 07.0.0, EB 70, Annex8)⁸
2. Tool 07 - Tool to calculate the emission factor for an electricity system (Version 07.0, EB 100, Annex 4)⁹
3. Tool 24 – Common practice (Version 03.1, EB 84, Annex 7)¹⁰
4. Tool 27- Investment analysis (Version 12.0, EB 112 annex 2)¹¹

3.2 Applicability of Methodology

The project activity is a green field grid connected floating solar power project with cumulative capacity of 140 MWp (100 MW_{AC}). Since the project is a large-scale project activity therefore Large-scale Consolidated Methodology ACM0002: Grid-connected electricity generation from renewable sources applies to project activities that include retrofitting, rehabilitation (or refurbishment), replacement or capacity addition of an existing power plant or construction and operation of a Greenfield power plant is applied. The applicability of the methodology is established through assessment of the following conditions.

ACM0002 - Grid-connected electricity generation from renewable sources (Version 21.0)		
Para No.	Applicability Condition	Justification of eligibility
4	This methodology is applicable to grid-connected renewable energy power generation project activities that: a) Install a Greenfield power plant;	The project activity is installation of a greenfield floating solar power (renewable). Hence the project activity meets the applicability condition of the methodology.

⁷ <https://cdm.unfccc.int/UserManagement/FileStorage/ZPFJL01OU2RYC6N3HASIXV7K84QBG9>

⁸ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-01-v7.0.0.pdf>

⁹ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-07-v7.0.0.pdf>

¹⁰ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-24-v1.pdf>

¹¹ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-27-v12.pdf>

	b) Involve a capacity addition to (an) existing plant(s); c) Involve a retrofit of (an) existing operating plant(s)/unit(s); d) Involve a rehabilitation of (an) existing plant(s)/unit(s); or e) Involve a replacement of (an) existing plant(s)/unit(s).	
5	In case the project activity involves the integration of a BESS, the methodology is applicable to grid-connected renewable energy power generation project activities that: a) Integrate BESS with a Greenfield power plant; b) Integrate a BESS together with implementing a capacity addition to (an) existing solar photovoltaic¹ or wind power plant(s)/unit(s); c) Integrate a BESS to (an) existing solar photovoltaic or wind power plant(s)/unit(s) without implementing any other changes to the existing plant(s); d) Integrate a BESS together with implementing a retrofit of (an) existing solar photovoltaic or wind power plant(s)/unit(s).	The project activity does not involve the integration of BESS hence this condition is not applicable.
6	The methodology is applicable under the following conditions: a) Hydro power plant/unit with or without reservoir, wind power plant/unit, geothermal power plant/unit, solar power plant/unit, wave power plant/unit or tidal power plant/unit; b) In the case of capacity additions, retrofits, rehabilitations or	The project activity involves construction and operation of greenfield grid-connected Floating solar power project and hence complies to the applicability condition 6.a of the methodology. Since the project activity does not include capacity additions, retrofits, rehabilitations, or replacements of existing plant/unit the applicability condition 6.b is not applicable/relevant for the project activity.

	replacements (except for wind, solar, wave or tidal power capacity addition projects) the existing plant/unit started commercial operation prior to the start of a minimum historical reference period of five years, used for the calculation of baseline emissions and defined in the baseline emission section, and no capacity expansion, retrofit, or rehabilitation of the plant/unit has been undertaken between the start of this minimum historical reference period and the implementation of the project activity; c) In case of Greenfield project activities applicable under paragraph 5 (a) above, the project participants shall demonstrate that the BESS was an integral part of the design of the renewable energy project activity (e.g. by referring to feasibility studies or investment decision documents); d) The BESS should be charged with electricity generated from the associated renewable energy power plant(s). Only during exigencies 2 may the BESS be charged with electricity from the grid or a fossil fuel electricity generator. In such cases, the corresponding GHG emissions shall be accounted for as project emissions following the requirements under section 5.4.4 below. The charging using the grid or using fossil fuel electricity generator should not amount to more than 2 per cent of the	The project activity does not involve the integration of BESS hence this condition 6.c is not applicable. The project activity does not involve the integration of BESS hence this condition 6.d is not applicable.
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	electricity generated by the project renewable energy plant during a monitoring period. During the time periods (e.g. week(s), months(s)) when the BESS consumes more than 2 per cent of the electricity for charging, the project participant shall not be entitled to issuance of the certified emission reductions for the concerned periods of the monitoring period.	
7	In case of hydro power plants, one of the following conditions shall apply: a) The project activity is implemented in existing single or multiple reservoirs, with no change in the volume of any of the reservoirs; b) The project activity is implemented in existing single or multiple reservoirs, where the volume of the reservoir(s) is increased and the power density, is greater than 4 W/m² c) The project activity results in new single or multiple reservoirs and the power density is greater than 4 W/m² d) The project activity is an integrated hydro power project involving multiple reservoirs, where the power density for any of the reservoirs, is lower than or equal to 4 W/m².	The project activity involves construction and operation of greenfield grid-connected floating solar power project using solar energy for generation of electricity hence the applicability condition “7” is not applicable/relevant to the project activity as the applicability conditions is related to hydro power projects
8	In the case of integrated hydro power projects, project participants shall: a) Demonstrate that water flow from upstream power plants/units spill directly to the downstream reservoir and that collectively constitute to the generation capacity of the	The project activity involves construction and operation of greenfield grid-connected floating solar power project using solar energy for generation of electricity, hence the applicability condition “8” is not applicable/relevant to the project activity as the applicability conditions is related to hydro power projects.

	integrated hydro power project; or b) Provide an analysis of the water balance covering the water fed to power units, with all possible combinations of reservoirs and without the construction of reservoirs.	
9	The methodology is not applicable to: a) Project activities that involve switching from fossil fuels to renewable energy sources at the site of the project activity, since in this case the baseline may be the continued use of fossil fuels at the site; b) Biomass fired power plants/units.	The project activity involves construction and operation of greenfield grid-connected floating solar power project using solar energy for generation of electricity hence the applicability condition "9" is not relevant as the same pertains to switching from fossil fuels to renewable energy sources or biomass fired power plants/units.
10	In the case of retrofits, rehabilitations, replacements, or capacity additions, this methodology is only applicable if the most plausible baseline scenario, as a result of the identification of baseline scenario, is "the continuation of the current situation, that is to use the power generation equipment that was already in use prior to the implementation of the project activity and undertaking business as usual maintenance"	The project activity involves construction and operation of greenfield grid-connected solar power project using solar energy for generation of electricity hence the applicability condition "10" is not relevant as the same pertains to retrofits, rehabilitations, replacements, or capacity additions.
11	In addition, the applicability conditions included in the tools referred to below apply	The applicability of the tools is outlined below

TOOL 01: Tool for the demonstration and assessment of additionality (Version 7.0.0)	
Applicability condition	Justification of eligibility
The use of the "Tool for the demonstration and assessment of additionality" is not mandatory for project participants when proposing new methodologies. Project participants may propose	Project participant is not proposing new methodologies.

alternative methods to demonstrate additionality for consideration by the Executive Board. They may also submit revisions to approved methodologies using the additionality tool.	
Once the additionally tool is included in an approved methodology, its application by project participants using this methodology is mandatory.	Project activity is applying additionality tool.

TOOL 7: Tool to calculate the emission factor for an electricity system (Version 7.0.0)	
Applicability Conditions	Justification for eligibility
This tool may be applied to estimate the OM, BM and/or CM when calculating baseline emissions for a project activity that substitutes grid electricity that is where a project activity supplies electricity to a grid or a project activity that results in savings of electricity that would have been provided by the grid (e.g. demand-side energy efficiency projects).	Since the project activity is grid connected, the condition is applicable and the OM, BM and/or CM has been calculated accordingly.
Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants only or, as an option, can include off-grid power plants. In the latter case, two sub-options under the step 2 of the tool are available to the project participants, i.e. option IIa and option IIb. If option IIa is chosen, the conditions specified in “Appendix 1: Procedures related to off-grid power generation” should be met. Namely, the total capacity of off-grid power plants (in MW) should be at least 10 per cent of the total capacity of grid power plants in the electricity system; or the total electricity generation by off-grid power plants (in MWh) should be at least 10 per cent of the total electricity	Since the project activity is grid connected, the condition is applicable and emission factor has been calculated accordingly.

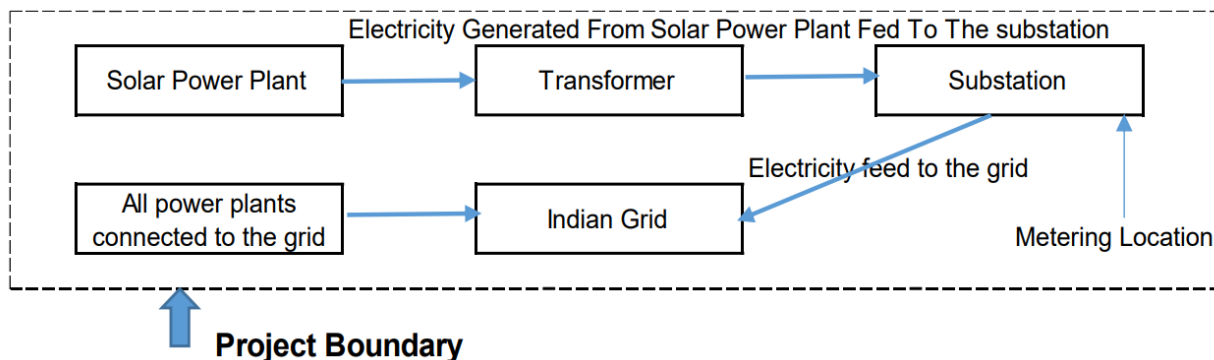
generation by grid power plants in the electricity system; and that factors which negatively affect the reliability and stability of the grid are primarily due to constraints in generation and not to other aspects such as transmission capacity.	
In case of CDM projects the tool is not applicable if the project electricity system is located partially or totally in an Annex I country.	The project activity is located in India, a Non-Annex I country. Therefore, this condition is not applicable to the project activity.
Under this tool, the value applied to the CO ₂ emission factor of biofuels is zero.	The project activity is a green field solar power plant and hence the condition of biofuel emission factor is not applicable.

Tool 24: Common practice (Version 03.1)	
Applicability Conditions	Justification of eligibility
This methodological tool is applicable to project activities that apply the methodological tool “Tool for the demonstration and assessment of additionality”, the methodological tool “Combined tool to identify the baseline scenario and demonstrate additionality”, or baseline and monitoring methodologies that use the common practice test for the demonstration of additionality.	The proposed project activity is applying the methodological tool “Tool for the demonstration and assessment of additionality”, the methodological tool “Combined tool to identify the baseline scenario and demonstrate additionality”, or baseline and monitoring methodologies that use the common practice test for the demonstration of additionality.
In case the applied approved baseline and monitoring methodology defines approaches for the conduction of the common practice test that are different from those described in this methodological tool, the requirements contained in the methodology shall prevail.	The applied approved baseline and monitoring methodology dose not define approaches for the conduction of the common practice test.

Tool 27: Investment analysis (Version 12.0)	
Applicability Condition	Justification of eligibility
Para 2 - This methodological tool is applicable to project activities that apply the methodological tool “Tool for the demonstration and assessment of additionality”, the methodological tool “Combined tool to identify the baseline scenario and demonstrate additionality”, the guidelines “Non-binding best practice examples to demonstrate additionality for SSC project activities”, or baseline and monitoring methodologies that use the investment analysis for the demonstration of additionality and/or the identification of the baseline scenario.	Proposed project activity applies Tool for the demonstration and assessment of additionality.
Para 3 - In case the applied approved baseline and monitoring methodology contains requirements for the investment analysis that are different from those described in this methodological tool, the requirements contained in the methodology shall prevail.	The applied approved baseline and monitoring methodology contains requirements for the investment analysis that are not different from those described in this methodological tool.

3.3 Project Boundary

The spatial extent of the project boundary includes the project power plant / unit and all power plants / units connected physically to the electricity system that the project power plant is connected to. The project boundary therefore includes the project site where the power plant has been installed, associated power evacuation infrastructure, energy metering points, switch yards the national grid of India.



The project boundary includes CO₂e as greenhouse gas (GHG) that would have emanated from the operation of grid connected fossil fuel-based power plants in the absence of the project activity.

Source		Gas	Included?	Justification/Explanation
Baseline	CO ₂ emissions from electricity generation in fossil fuel fired power plants that are displaced due to project activity	CO ₂	Yes	Main emission source
		CH ₄	No	Minor emission source
		N ₂ O	No	Not Applicable
Project	Solar energy under the project activity	CO ₂	No	As the project activity is a zero-emission grid connected solar power project no emission will result from the project activity in operation phase.
		CH ₄	No	As the project activity is a zero-emission grid connected solar power project no emission will result from the project activity in operation phase.
		N ₂ O	No	As the project activity is a zero-emission grid connected solar power project no emission will result from the project activity in operation phase.

3.4 Baseline Scenario

As per the approved consolidated Methodology ACM0002 (Version 21.0), baseline scenario would be defined as: “If the project activity is the installation of a Greenfield power plant, the baseline scenario is electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in the “Tool to calculate the emission factor for an electricity system (Version 07.0)”.

Hence, baseline for the project activity is the equivalent amount of power from the INDIAN grid.

3.5 Additionality

The additionality of the Project shall be demonstrated by applying the following approach, consisting of two components:

1. A Legal Requirement Test; and
2. An Additionality Test either based on a Positive List test or a projects-specific additionality test.

The project is not mandated/enforced by law and is entirely a voluntary activity. Since voluntary commitments/agreements within a sector or by an entity do not constitute the legal requirement.

Specify the methodology, activity requirement or product requirement that establishes deemed additionality for the proposed project (including the version number and the specific paragraph, if applicable).	This project applies an approved large-scale methodology ACM0002 “Consolidated baseline methodology for grid connected electricity generation from renewable sources”, Version 21.0. Selected methodology has been applied together with the “tool to calculate the emission factor for an electricity system, version 7.0” and “tool for assessment and demonstration of additionality, version 7.0.0”. These are the latest version of the methodology and related additionality & emission factor calculation tool.
Describe how the proposed project meets the criteria for deemed additionality.	Project revenue is not financially attractive as discussed in investment analysis section below (benchmark and sensitivity analysis). Continuation of the current situation supply of equal amount of electricity by the newly built grid connected power plants. Continuation of the current situation is not considered as a realistic alternative due to increasing electricity demand therefore new power plants should be constructed which includes mainly thermal power plants. Implementation of the project is additional to the baseline scenario

	which is an alternative 2 above and therefore reduces the emissions. The project activity comes under white category as per local regulation, thus there shall be no necessity of obtaining the Consent to Operate” for White category of industries. Since project activity falls under white category and the non-polluting nature of project fulfils the compliance to the local laws and regulations. The Project activity conforms to all the applicable laws and regulations in India: 1. Power generation using renewable energy is not a legal requirement or a mandatory option¹². 2. There are state and sectoral policies, framed primarily to encourage renewable power projects. 3. These policies have also been drafted realizing the extent of risks involved in the projects and to attract private investments. 4. The Indian Electricity Act, 2003 (May 2007 Amendment) does not influence the choice of fuel used for power generation. 5. There is no legal requirement on the choice of a particular technology for power generation The both alternatives are in compliance with laws and regulations required. There is no mandatory requirement to implement the project activity. In accordance with common practice analysis there is no plants similar to the proposed project and built without carbon revenue, the proposed type of project should not be considered as a common practice. Hence, project is additional in this aspect.
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¹² Electricity Act, 2003 and its subsequent amendments

Additionality Assessment for large scale project activity Instances

The present project generates power using solar energy which is a renewable, zero emission source of energy. However, the project activity does not comply to the criteria of positive list as provided in CDM Tool 32: “Methodological Tool – Positive List of Technologies” and hence additionality of the project activity is demonstrated through a project specific additionality test.

Baseline considerations for the project are based on approved consolidated baseline methodology. ACM0002 (Version 21.0). The methodology requires the project investor to determine the additionality based on “Methodological Tool- Tool for the demonstration and assessment of additionality”, Version 7.0.0. The tool provides a step-wise approach to demonstrate and assess the additionality of a project (figure below). These steps are:

- a) Step 0 Demonstration whether the proposed project activity is the first-of-its-kind;
- b) Step 1 Identification of alternatives to the project activity;
- c) Step 2 Investment analysis;
- d) Step 3 Barriers analysis; and
- e) Step 4 Common practice analysis

The step-wise approach to establish additionality of the project activity has been followed, details of which are provided in the following paragraphs:

Step 0: Demonstration whether the proposed project activity is the first-of-its-kind

The project activity is a large-scale solar power project undertaken in Indian state of Madhya Pradesh. This is not the first such project to be installed in the country or in the state and therefore project activity does not meet this criterion.

Step 1: Identification of alternatives to the project activity consistent with current laws and regulations

Sub-step 1a: Define alternatives to the project activity: Identify realistic and credible alternative(s) available to the project participants or similar project developers that provide outputs or services comparable with the proposed project activity.

The purpose of the project activity is aimed to generate power from the solar power plant using renewable energy (solar) resources and feed the electricity generated to the grid. Hence, the following alternatives are considered:

Alternative 1: The proposed project activity undertaken without being registered as a Proposed project activity.

The option of proposed project activity being undertaken without being registered as a project activity is in compliance with all applicable legal and regulatory requirements and can be a part of the baseline. However, this is not a plausible option as deployment of the project activity faces significant financial barriers established in the subsequent section. The electricity generated from the project activity would have been fed to the Fertilizer plant through India's National Grid, which has a number of power plants including existing and upcoming ones which would be used in the absence of project activity. In the absence of project activity, power from the same grid will continue to provide for the requirement of Plant and the country.

Alternative 2: Continuation of the current situation (No proposed project activity and equivalent amount of energy would have been produced by the grid electricity system through its currently running power plants and by new capacity addition to the grid).

In such case the project owner would have continued without investment in project activity and continued with its usual business activities. In such case grid would continue with the fossil fuel-based power projects and this would result in GHG emissions. Hence, existing power plant and the new capacity add-on from a fossil fuel-based power plant is appropriate, realistic & credible baseline alternative for the project activity.

Outcome of Sub-step 1a: Identified realistic and credible alternative scenario (s) to the project activity.

Of the two alternatives outlined above, the first alternative is not possible as project activity is not viable without carbon credit benefits and second alternative is the baseline scenario for the project activity as per methodology. Therefore, continuation of the current situation is the likely alternative to the project activity.

The project being a green field project activity the baseline scenario in line with the methodology is "the baseline scenario is electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in the "Tool to calculate the emission factor for an electricity system Version 7.0"

Sub-step 1b: Consistency with mandatory laws and regulations: The alternative(s) shall be in compliance with all applicable legal and regulatory requirements, even if these laws and regulations have objectives other than GHG reductions, e.g. to mitigate local air pollution.

Installation of a large-scale solar power project including the one proposed as project activity is consistent with mandatory laws and regulations at national (India) and sub-national level (state of Madhya Pradesh). The environmental regulations, legislations and policy guidelines in respect to the project activity is governed by Ministry of Environment, Forest and Climate Change (MoEF&CC), Delhi supported by Central Pollution Control Board (CPCB). In accordance to

Annexure-II MOEF&CC, OM on J-11013/41/2006-IA. II (I) dated on 07-july-2017 Solar Photovoltaic Power Projects are not covered under the ambit of EIA Notification, 2006 and its subsequent amendments and hence, it does not require preparation of EIA and pursuing Environmental Clearance from MoEF&CC.

Moreover, Solar Power Project is considered under “white category” for Consent to Establish/Operate as per regulation, and thus there is no necessity of obtaining the Consent to Operate”. Since project activity falls under white category and the non-polluting in nature, the project fulfils the compliance to the local environmental laws and regulations.

Overall, the project activity is consistent with the following mandatory laws and regulations.

- Electricity Act, 2003 and it's subsequent amendment as it does not influence the choice of fuel used for power generation nor has it mandated power generation using renewable energy only.
- Environmental (Protection) Act, 1986 and amendment(s)
- Environmental Impact Assessment (EIA) Notification, 2006 and amendment(s)
- The Air (Prevention and Control of Pollution) Act, 1981 including Rules 1982 and 1983 and amendment(s)
- The Water Prevention and Control of Pollution), Cess Act, 1977 including Rules 1978 and 1991 and amended in March, 2003
- Noise Pollution (Regulation and Control) Amendment Rules, 2017
- Hazardous and other Wastes (Management & Transboundary Movement) Rules, 2016 and amended in November, 2021
- E-waste (Management and Handling) Rules, 2020
- Bio-Medical Waste (Management and Handling) Rules 2016
- Plastics Waste Management Rules, 2016 amended in 2021
- Batteries (management and handling) rules 2019

Outcome of Sub-step 1b: Identified realistic and credible alternative scenario(s) to the project activity that are in compliance with mandatory legislation and regulations taking into account the enforcement in the region or country and EB decisions on national and/or sectoral policies and regulations.

Hence, both the alternatives enlisted above are found to comply with the mandatory laws and regulations taking into account the enforcement of the legislations in the region or country and EB decisions on national and/or sectoral policies and regulations. However, Alternative 2 “Continuation of the current situation” has been selected as the appropriate baseline alternative for this project activity in line with methodology.

Step 2: Investment analysis

Investment analysis is carried out to determine on whether the proposed project activity is economically or financially less attractive than at least one other alternative, identified in step 1, without the revenue from the sale of emission reductions credits. This is demonstrated in line with the Tools for sections as per “Investment Analysis” (Ver 12.0).

Sub-step 2a: Determine appropriate analysis method

There are three options for the determination of analysis method which are:

- I. Simple Cost Analysis
- II. Investment Comparison Analysis and
- III. Benchmark Analysis

The Project activity envisages to export the power to Indian grid and the revenues from the sale of electricity would be generated in accordance with the terms and tariffs established in the Power Purchase Agreement (PPA). Thus, simple cost analysis (Option I) cannot be used as the analysis method as the sale of the units of generated electricity shall result in a revenue stream during the operations of the Project activity.

In the absence of the project activity grid electricity would have been the obvious choice for the Project which requires no investment. Hence investment comparison analysis (Option II) is also not appropriate for the project activity.

After eliminating Option, I and Option II, the use of Benchmark analysis (Option III) is the method of analysis that has been selected as the most suitable method. This method determines the attractiveness of the project activity for the investors, as well as provides a measure of the viability of the investment to generate revenues during its operation, as compared with other avenues and investment options. Hence, the Benchmark analysis method is to be employed for analysis of the said project.

Sub-step 2b (Option III): Apply benchmark analysis

The investment analysis using Benchmark analysis approach (Option III) has been chosen. Further, this method illustrates that the evaluation of the project by the project owner was carried out before the decision to undertake the project was taken and management approval had been granted.

Choice of Financial Indicator:

According to the “Tool for demonstration and assessment of Additionality”, Version 07.0.0, the financial indicator can be based either on (1) project IRR or (2) equity IRR. There is no general preference between the approaches (1) or (2). The benchmark chosen for analysis shall be fully consistent with the choice of approach. Therefore, in accordance with the guidance, the relevant financial indicator for project activity has been chosen as post tax equity IRR.

Default Value Benchmark:

The cost of equity is determined by selecting the values provided in the Appendix, i.e., Default values for cost of equity (expected return on equity) is presented below:

Appendix of the CDM TOOL 27: Investment Analysis Version 12.0 specifies default value of expected return on equity in real terms for Energy Industries (Group 1) in India = 9.77%

The Required return on equity (benchmark) was computed in the following manner:

$$\text{Nominal Benchmark} = \{(1 + \text{Real Benchmark}) * (1 + \text{Inflation rate})\} - 1$$

Where:

Default value for Real Benchmark = 10.55% (as per Appendix of CDM TOOL 27: Investment Analysis).

Inflation Rate forecast for by Reserve Bank of India (RBI) (i.e. Central Bank of India) for India.

Benchmark estimation:

Benchmark for Floating Solar project of amp Energy

Description	Value	Source Link
Default Value for India as per UNFCCC guidelines	09.77%	EB 116, Annex 2, TOOL27 "Investment Analysis", Version 12.0 (Web Link - https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-27-v12.pdf)
Inflation forecast as per RBI for 10yr	4.00%	https://rbi.org.in/home.aspx
Benchmark	14.97%	Calculated

Sub-step 2c: Calculation and comparison of financial indicators (only applicable to Options II and III): The Post tax Equity IRR is evaluated for the entire lifetime of the project activity, i.e. 25 years. It is calculated based on the cash outflows from and cash inflows into the project activity. substantiates that the investment is not financially attractive (Equity IRR for the project activity is less than the Benchmark). Thus, it can be easily concluded that project activity is additional & is not business as usual scenario.

The input assumption referred below:

Details of the project		Source	Link
Project Location	Madhya Pradesh	DPR	
Total Capacity (MWac)	100.00	DPR	
Expected Date of Commissioning	31-01-2024	DPR	
Life of the plant (Yrs.)	25	DPR	-
Generation of electricity			
PLF (%)	24.32%	PV Syst Report @ P90 level	-
Annual generation (kWh)	213,043,200	Calculated Value	
Annual Degradation per year from 2nd Yr onwards	0.30%	DPR	-
Tariff rate at the decision making (INR/kWh)	3.19	DPR	-
Operation and maintenance cost and Insurance			
O & M Expenses (INR Mn.)	25.96	DPR	-
O & M free for (Yr.)	-		-
Escalation in the operational expenses (%)	2.00%	DPR	-
Management Fee	0.00		-
Escalation (%)	4.00%	DPR	-
Admin Fee (₹ mn)	0.35	DPR	-
Escalation (%)	4.00%	DPR	-
Salary (₹ mn)	0.35	DPR	-
Escalation (%)	4.00%	DPR	-
Lender maintenance (₹ mn)	0.45	DPR	-
Escalation (%)	4.00%	DPR	-
Insurance (₹ mn)	8.85	DPR	-
Escalation (%)	0.00%		-
Any other cost	0.00		-
Financial parameters			
TOTAL COST (INR Mn.)	6,000.00	DPR	
Loan Amount (INR Mn.)	4,500.00	DPR	
Equity Investment (INR Mn.)	1,500.00	Calculated Amount	

Term loan			
Loan Amount (INR Mn.)	4,500.00	75:25 ratio as per DPR	-
Interest rate (%)	9.15%	DPR	-
Loan Tenure (Qtr.)	80.00	DPR	-
Moratorium Period (Qtr.)	4.00		-
Repayment Period (Qtr.)	76.00	Calculated Value	-
Repayment instalments value (INR Mn.)	59.21	Calculated Value	-
1st instalment from (Qtr. end)	31-Mar-25	Considered from the next Quarter End	-
Book Depreciation (SLM Method)			
Gross Depreciable Value (INR Mn.)	6,000.00	Calculated Value	
Salvage Value (%)	10.00%	DPR	-
Salvage value (INR Mn.)	600.00	Calculated Value	
Net Depreciable Value (INR Mn.)	5,400.00	Calculated Value	
Residual Value (INR Mn.)	600.00	Calculated Value	-
IT Depreciation			
IT Depreciation (%)	7.69%	As Per Depreciation Rates for Power Generating Units	https://incometaxindia.gov.in/charts%20%20tables/depreciation%20rates.htm
Income Tax			
Financial Year	FY 2023-24		
Income tax rate (%)	30.00%	Tax rated applicable @ FY 2020-21 for a domestic company	https://incometaxindia.gov.in/Books%20%20Pamphlets/e-Pdf-INCOME-TAX-RATES-english-2022.pdf
MAT (%)	15.00%		
Surcharge (%)	7.00%		
Health & Education cess (%)	4.00%		
Service tax (%)	18.00%	As Per Income Tax Rule	https://old.cbic.gov.in/resources//htdocs-cbec/gst/Work%20Contracts%20in%20GST.pdf
Final Tax rates			
Income tax rate (%)	33.38%	Calculated Value	
MAT (%)	16.69%	Calculated Value	
Service tax (%)	18.72%	Calculated Value	

Outcome of Step 2:

Calculation of equity IRR is submitted in the form of excel sheet.

Equity IRR	Benchmark (Equity IRR)
6.42 %	14.16%

This substantiates that the investment is not financially attractive (Equity IRR for the project activity is less than the Benchmark Equity IRR). Thus, it can be easily concluded that project activity is additional & is not business as usual scenario.

Sub-step 2d: Sensitivity Analysis

Addressing Guidance under Sensitivity Analysis Version-12.0, following factors has been subjected to sensitivity analysis:

1. PLF
2. O&M Cost
3. Project Cost
4. Tariff

The rationale of sensitivity is, "The ultimate objective of the sensitivity analysis is to determine the likelihood of the occurrence of a scenario other than the scenario presented, in order to provide a cross-check on the suitability of the assumptions used in the development of the investment analysis."

Sensitivity Analysis

The results of sensitivity analysis show that the even with a variation of +10% & -10% in the project cost, O&M cost, PLF and Tariff Rate Equity IRR is significantly lower than the benchmark. And it is evident from the results given above; the project remains additional even under the most favourable conditions.

Probability to breach the benchmark:
Sensitivity Parameter 1: PLF or generation
The PLF or generation is considered at the 90% probability level for entire lifetime of the project activity. Hence, there is no probability to get variation in the same. However, Sensitivity is carried out for +/-10% even then the benchmark is not breached
Sensitivity Parameter 2: O&M
The sensitivity analysis reveals that O&M will breach the benchmark at negative values and is hypothetical case. Since the O&M cost is generally subject to escalation and also subject to inflationary pressure, any reduction in the O&M costs is highly unlikely. Hence, the reduction in the O&M cost is highly unlikely.
Sensitivity Parameter 3: Project Cost

Project Cost for financial analysis is considered based on the information of DPR, being available at the time of investment decision making to go ahead with the project activity. However, the Sensitivity is carried out for 10% variation and for threshold level below which benchmark is not breached. Thus, it is unlikely that project cost will change beyond sensitivity range

Sensitivity Parameter 4: Tariff Rate

The tariff is fixed for entire lifetime of the project activity. Hence, there is no probability to get variation for the same. However, Sensitivity is carried out for +/-10% even then the benchmark is not breached

Result of Sensitivity Analysis Table:

Parameters	-10%	IRR (Base Case)	+10%	IRR Breaching Value
PLF or Generation	3.28%	6.42%	9.81%	22.27%
O&M Cost	6.62%	6.42%	6.23%	-385.00%
Project Cost	9.76%	6.42%	3.81%	-20.15%
Tariff Rate	3.28%	6.42%	9.81%	22.27%

Step 3: Barrier Analysis

As per Tool for demonstration and assessment of additionality" (Version 07.0.0), Step 2 or Step 3 or both can be used to demonstrate additionality of the project activity. In this case, Step 3 is not being used for the purpose.

Step 4: Common Practice Analysis

As per para 57 of Tool for demonstration and assessment of additionality" (Version 07.0.0), Step 2 analysis shall be complemented with an analysis of extent to which the proposed project type (e.g., technology or practice) has already diffused in the relevant sector and region. This test is a credibility check to complement the investment analysis.

Common Practice Analysis – 100 MW Floating Solar Power Project by AMP Energy In Madhya Pradesh

Step (1): Calculate applicable capacity or output range as +/-50% of the total design capacity or output of the proposed project activity.

Range	Capacity	Unit
+50%	150	MW
Capacity of the proposed project activity	100	MW

-50%	50	MW
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Step (2): Identify similar projects (both CDM and non-CDM) which fulfil all of the following conditions:

- The projects are located in the applicable geographical area;
- The projects apply the same measure as the proposed project activity;
- The projects use the same energy source/fuel and feedstock as the proposed project activity, if a technology switch measure is implemented by the proposed project activity;
- The plants in which the projects are implemented produce goods or services with comparable quality, properties and applications areas (e.g., clinker) as the proposed project plant;
- The capacity or output of the projects is within the applicable capacity range for the chosen projects.
- The projects started commercial operation before the PSF is published for global stakeholder consultation or before the start date of proposed project activity, whichever is earlier for the proposed project activity.

Identification of the similar projects (CDM and non-CDM) is carried out as per sub-steps of Step (2) as follows:

- As the project is in Madhya Pradesh state of India, therefore, projects in the geographical area of Madhya Pradesh have been chosen for analysis. The project activity involves generation of electricity from solar energy. The project activity is in the states of Madhya Pradesh in India and the policy applicable for the solar projects is regulated by respective state policy. The policies/tariff for each state is regulated by State Electricity Regulatory Commissions of respective states and they differ for respective states. The project is Floating PV solar and a 25% equity-based project and generated power will be exported to grid. Hence, all state electricity regulations are applicable for this project. So, solar projects, which are exporting electricity to grid are followed different state electricity regulatory policies.
- The project activity is a green-field solar power project and uses measure (b) “Switch of technology with or without change of energy source including energy efficiency improvement as well as use of renewable energies”. Therefore, projects applying same measure (b) are candidates for similar projects.
- The energy source used by the project activity is solar. Hence, only solar energy projects have been considered for analysis.

- The project activity produces electricity; therefore, all power plants that produce electricity are candidates for similar projects.
- The capacity range of the projects is within the applicable capacity range from 50 MW to 150 MW.
- Power Purchase Agreement date of the project activity is 04/08/2022, therefore, projects, which have started commercial operation before 04/08/2022, have been considered for analysis.

Numbers of similar projects identified, which fulfil above-mentioned conditioned are $N_{\text{solar}} = 0$

Step (3): Within the projects identified in Step 2, identify those that are neither registered CDM project activities, project activities submitted for registration, nor project activities undergoing validation. Note their number, N_{all} .

Project activities, which have got registered or are under validation with CDM/VCS/GS/GCC have been excluded in this step. The list of the power plants identified is provided to the Verifier. After excluding the registered and under validation projects the total number of projects.

$N_{\text{all}} = 0$

Step (4): Within similar projects identified in Step 3, identify those that apply technologies that are different to the technology applied in the proposed project activity. Note their number N_{diff} .

As per the tool on Common Practice, the project activities have been separated from the different technologies on the basis two criteria:

Different technologies - are technologies that deliver the same output and differ by at least one of the following (as appropriate in the context of the measure applied in the proposed project activity and applicable geographical area):

(a) Energy source/fuel (example: energy generation by different energy sources such as wind and hydro and different types of fuels such as biomass and natural gas);

(b) Feed stock (example: production of fuel ethanol from different feed stocks such as sugar cane and starch, production of cement with varying percentage of alternative fuels or less carbon - intensive fuels);

(c) Size of installation (power capacity)/energy savings:

(i) Micro (as defined in paragraph 24 of decision 2/CMP.5 and paragraph 39 of decision 3/CMP.6); (ii) Small (as defined in paragraph 28 of decision 1/CMP.2); (iii) Large.

(d) Investment climate on the date of the investment decision, inter alia:

- (i) Access to technology;
- (ii) Subsidies or other financial flows;
- (iii) Promotional policies;
- (iv) Legal regulations.

(e) Other features, inter alia:

- (i) Nature of the investment (example: unit cost of capacity or output is considered different if the costs differ by at least 20 %).

The Project is 30% equity-based and Floating PV solar Captive project and deployed for export of power after paying wheeling charges. Hence, the investment climate of the project is different from any other project of different tariff or revenue structure. So solar power project that are not covered under reverse bid project are considered as different from the project activity.

Hence, projects where either of the conditions is satisfied those projects are counted for calculating N_{diff} projects.

$$N_{diff} = 0$$

Step (5): calculate factor $F = 1 - N_{diff}/N_{all}$ representing the share of similar projects (penetration rate of the measure/technology) using a measure/technology similar to the measure/technology used in the proposed project activity that deliver the same output or capacity as the proposed project activity.

The factor for the project activity is calculated as follows:

In terms of the formula: $F = 1 - N_{diff}/N_{all}$, the result is mathematically meaningless.

$$N_{all} - N_{diff} = 0$$

Thus, the project does not fulfil the requirement that F is greater than 0.2 and $N_{all} - N_{diff}$ is greater than 3, and the project is not a common practice

3.6 Methodology Deviations

There is no deviation in methodology applied by the project activity hence not applicable.

4 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

The baseline emissions are the product of generate power from the solar power plant baseline $EG_{PJ,Y}$ expressed in MWh of electricity produced by the renewable generating unit multiplied by an emission factor. As per Methodology ACM0002 Ver-21.0 the baseline emission equations are used.

$$BE_y = EG_{PJ,Y} * EF_{Grid,CM,y}$$

Where:

BE_y = Baseline emissions in year y (t CO₂/yr)

$EG_{PJ,Y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh/yr)

$EF_{Grid,CM,y}$ = Combined margin CO₂ emission factor for grid connected power generation in year y calculated using TOOL07 (t CO₂/MWh)

For Green field project

$$EG_{PJ,y} = EG_{Facility,y}$$

Where;

$EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh/yr)

$EG_{Facility,y}$ = Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh/yr)

$$BE_y = 205,560 * 0.9310$$

$$BE_y = 191,376 \text{ tCO}_2$$

4.2 Project Emissions

As per para 31 of the applied methodology ACM0002 (version 21.0), no project emissions are considered in this project activity.

Hence, project emissions $PE_y = 0 \text{ tCO}_2\text{e}$.

4.3 Leakage

As per Para 61 of ACM0002 Version 21.0 The emissions potentially arising due to activities such as power plant construction and upstream emissions from fossil fuel use (e.g. extraction, processing, transport etc. are neglected. No other emissions are considered.

Hence, $LE_y = 0$.

4.4 Net GHG Emission Reductions and Removals

Year	Estimated baseline	Estimated project emissions or removals (tCO ₂ e)	Estimated leakage	Estimated net GHG emission
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	emissions or removals (tCO ₂ e)		emissions (tCO ₂ e)	reductions or removals (tCO ₂ e)
Year 1	198,343	0	0	198,343
Year 2	197,450	0	0	197,450
Year 3	195,673	0	0	195,673
Year 4	193,031	0	0	193,031
Year 5	189,557	0	0	189,557
Year 6	185,292	0	0	185,292
Year 7	180,289	0	0	180,289
Total	1,339,635	0	0	1,339,635

5 MONITORING

5.1 Data and Parameters Available at Validation

Data / Parameter	EF _{Grid,OM,y}
Data unit	tCO ₂ /MWh
Description	Operating Margin CO ₂ emission factor in the year y
Source of data	CO ₂ Emission Database, Version 18.0, December 2022 published by Central Electricity Authority (CEA), Government of India.
Value applied	0.9518
Justification of choice of data or description of measurement methods and procedures applied	Not Applicable
Purpose of Data	Calculation of baseline emissions
Comments	This parameter is fixed ex-ante for the entire crediting period.

Data / Parameter	EF _{Grid,BM,y}
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Data unit	tCO ₂ /MWh
Description	Build margin CO ₂ emission factor I n the year y.
Source of data	CO ₂ Emission Database, Version 18.0, December 2022 published by Central Electricity Authority (CEA), Government of India.
Value applied	0.8687
Justification of choice of data or description of measurement methods and procedures applied	Not Applicable
Purpose of Data	Calculation of baseline emissions
Comments	This parameter is fixed ex-ante for the entire crediting period.

Data / Parameter	EF _{Grid,CM,y}
Data unit	tCO ₂ /MWh
Description	Combined margin CO ₂ emission factor in the year y
Source of data	CO ₂ Emission Database, Version 18.0, December 2022 published by Central Electricity Authority (CEA), Government of India.
Value applied	0.9310
Justification of choice of data or description of measurement methods and procedures applied	Not Applicable
Purpose of Data	Calculation of baseline emissions
Comments	This parameter is fixed ex-ante for the entire crediting period.

5.2 Data and Parameters Monitored

Data / Parameter	EG _{PJ,y}
Data unit	MWh/year

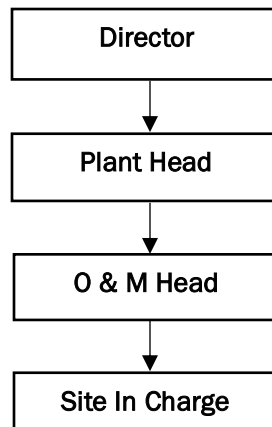
Description	Quantity of net electricity generation supplied by the project plant/unit to the grid in year y.
Source of data	Monthly Joint Meter Reading
Description of measurement methods and procedures to be applied	Not Applicable
Frequency of monitoring/recording	Monthly (Continuous measurement)
Value applied	205,560 MWh (Average annual generation for 7 years)
Monitoring equipment	Energy meters
QA/QC procedures to be applied	The meters are approved, tested & sealed by the State Utility and are in the custody of State Utility. The metering arrangement, accuracy class of meters, calibration frequency is under control of state utility and Project owner do not have any control on it. The calibration of all the meters is planned to be carried out in-line with the National standard which recommends at least once in 5-year calibration. Faulty meters will be duly replaced. The meters will be of accuracy class 0.2s. The electricity generation (gross or net) shall be cross-checked with records of electricity sale (e.g. sales invoice/receipt).
Purpose of data	Calculation of baseline emission
Calculation method	The Net electricity exported/supplied to the grid by the project activity is estimated as a difference of electricity exported to the grid and electricity imported from the grid. Net Electricity Generation = $EG_{\text{export to Grid}} - EG_{\text{Import from Grid}}$ The net electricity supplied/exported to the grid can be obtained from the record of Monthly Joint Meter Reading by the state utility
Comments	Data will be archived in paper & electronically for a period of 2 years beyond the end of crediting period or of the last issuance of credits for this project activity, whichever occurs later.

5.3 Monitoring Plan

The monitoring plan is developed in accordance with the modalities and procedures for CDM project activities and is proposed for grid-connected solar power project being implemented. The monitoring plan, which will be implemented by the project owner describes about the monitoring organisation, parameters to be monitored, monitoring practices, quality assurance, quality control procedures, data storage and archiving.

The authority and responsibility for registration, monitoring, measurement, reporting and reviewing of the data rests with the project participant. PO proposed the following structure for data monitoring, collection, data archiving and calibration of equipment's for this project activity.

Organizational Structure for Monitoring-



Responsibility of Head (Director): Tracking and reviewing the overall functioning and maintenance of the project activity from Plant Head (Operations). Plant Head (Operations) will be reporting Director.

Responsibilities of Plant Head: Overall functioning of the project activity and Coordinating with the O & M Manger and Team for the proper functioning of Project activity. He will be reporting to Director.

Responsibilities of O & M Head: O & M Manager and team is responsible for Operations and Maintenance related issues, they are also responsible for day-to-day data collection and monitoring, ensures completeness and reliability of data (calibration of equipment).

The Site In-charge will be responsible for carrying out internal auditing and QA/QC. All the values from generation record will be checked with JMR and invoices for consistency. In case there are any non-conformances identified. The Site In-charge will investigate the error and revise the

record to correct it. In any case where values have slightest of variation in different records the most conservative value will be taken in the project monitoring report.

Data Measurement

The export and import energy will be measured continuously using Main and Check meters located at the substations. Readings of meters shall be taken on monthly basis by authorized officer of utility in the presence of the representative of Project Owner (PO). Based on the Meter Reading Statement, invoices will be raised. These invoices can be used for cross checking the meter readings taken for the respective project activity.

Data Collection and Achieving

Readings from meters will be collected in the presence of the plant in-charge. Export and Import data would be recorded and stored in logs as well as in electronic form. The records are checked periodically by the Plant Manager and discussed thoroughly with the plant supervisor. The period of storage of the monitored data will be 2 years after the end of crediting period or till the last issuance of ACCs for the project activity whichever occurs later.

Emergency Preparedness

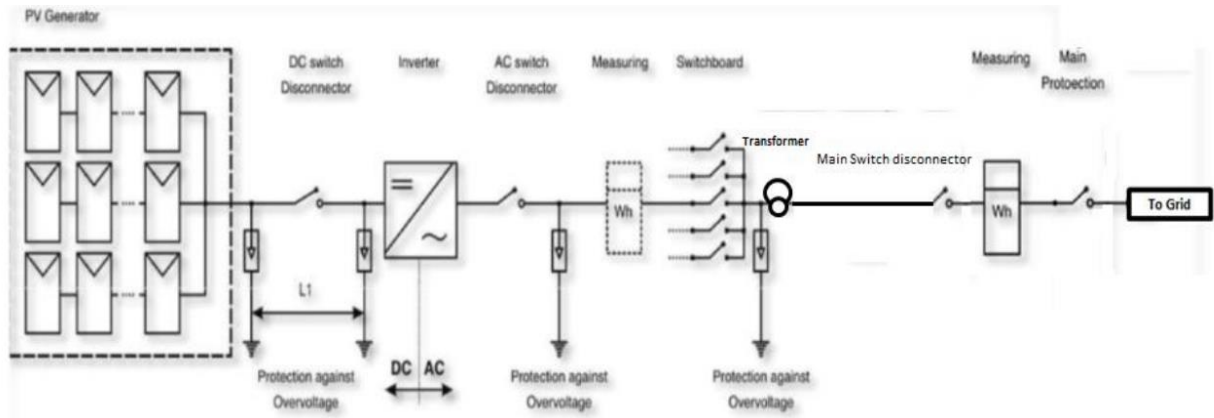
The project activity will not result in any unidentified activity that can result in substantial emissions from the project activity. No need for emergency preparedness in data monitoring is visualized. In the event that the main meter, which is used to record the net electricity exported by the project, is found to be faulty it will be repaired or replaced by authorized officer of SEB and the data from the check meter will be used in its place. In the unlikely event that the check meter fails it will also be repaired or replaced.

Personnel Training

In order to ensure proper functioning of the project activity and proper monitoring of emission reductions, the staff will be trained. The plant staff will be trained in equipment operation, data recording, reports writing, operation & maintenance, and emergency procedures in compliance with the monitoring plan.

Metering Arrangement

Line diagram with metering arrangement for the solar project activity is shown below:



QA/QC Procedures The energy meters at the feeders are maintained and owned by state electricity board. Neither the project owner nor the site personnel have any control over it. The records will be crosschecked with the records of sold electricity. The meters are calibrated by state electricity board out in-line with the Nation standard which recommends at least once in 5-year calibration or whenever abnormal difference/inconsistency is observed between main meter and check meter.